

# Synchronous convergence

## 1. Preliminary definitions

**Definition 1.1.** A set  $I$  together with a relation  $\leq$  is said to be *directed* by  $\leq$  if the following properties hold:

- **Reflexivity:**  $i \leq i$  for all  $i \in I$ ;
- **Transitivity:** If  $i_1 \leq i_2$  and  $i_2 \leq i_3$ , then  $i_1 \leq i_3$ ;
- **Upper bound property:** For all  $i_1, i_2 \in I$ , there is  $i \in I$  such that  $i_1 \leq i$  and  $i_2 \leq i$ .

**Definition 1.2.** Let  $I, J$  be two directed sets. A map  $\sigma : J \rightarrow I$  is said to be *directed* if for any  $i_0 \in I$  there is  $j_0 \in J$  such that  $j \geq j_0$  implies  $\sigma(j) \geq i_0$ .

**Definition 1.3.** Let  $X$  be a set and let  $I$  be a directed set. A function  $I \rightarrow X$  is called a *net* in  $X$ , denoted by  $(x_i)_{i \in I}$ . The set of all nets in  $X$  (with varying directed sets) will be denoted by  $\mathcal{N}(X)$ .

**Definition 1.4.** Let  $X$  be a topological space, and consider a net  $(x_i)_{i \in I}$  in  $X$ . We say that  $(x_i)_{i \in I}$  *converges* to  $x \in X$  and write  $x_i \xrightarrow{i \in I} x$ , if for any neighborhood  $U$  of  $x$ , there is  $i_0 \in I$  such that  $i \geq i_0$  implies  $x_i \in U$ . In other words,  $x_i$  is *eventually* in any neighborhood of  $x$ .

**Definition 1.5.** Let  $(x_i)_{i \in I}$  be a net in a set  $X$ . If  $\sigma : J \rightarrow I$  is a directed map, then the net  $(x_{\sigma(j)})_{j \in J}$  is called a *subnet* of  $(x_i)_{i \in I}$ .

**Definition 1.6.** Let  $X$  be a set together with a family  $\mathcal{C} \subset \mathcal{N}(X) \times X$ . Then  $(X, \mathcal{C})$  is called a *convergence space* if the following conditions are satisfied:

- For any  $x \in X$  and any directed set  $I$ , we have  $((x_i)_{i \in I}, x) \in \mathcal{C}$ . In other words, any constant net  $\mathcal{C}$  – converges to the constant;
- If  $((x_i)_{i \in I}, x_0) \in \mathcal{C}$ , then  $((x_{\sigma(j)})_{j \in J}, x_0) \in \mathcal{C}$  for any subnet  $(x_{\sigma(j)})_{j \in J}$  of  $(x_i)_{i \in I}$ .

The condition  $((x_i)_{i \in I}, x_0) \in \mathcal{C}$  may be denoted by  $x_i \xrightarrow{i \in I}^{\mathcal{C}} x_0$  or just  $x_i \xrightarrow{i \in I} x_0$ .

**Remark 1.1.** Any topological space  $X$  automatically receives a convergence space structure as per [Definition 1.4](#).

**Definition 1.7.** A function  $f : X \rightarrow Y$  between two convergence spaces is said to be *continuous* if for any net  $(x_i)_{i \in I}$  converging to  $x$  in  $X$ , the net  $(f(x_i))_{i \in I}$  converges to  $f(x)$  in  $Y$ .

**Definition 1.8.** Let  $X$  be a set together with a family  $\mathcal{C} \subset \mathcal{N}(X) \times X$ . Then the family  $\mathcal{C}$  is called a *convergence class* if the following hold:

- $(X, \mathcal{C})$  is a convergence space;
- Let  $x \in X$ . If every subnet of  $(x_i)_{i \in I} \subset X$  has a subnet that  $\mathcal{C}$ -converges to  $x$ , then  $x_i \xrightarrow{i \in I}^{\mathcal{C}} x$ .
- Consider a directed set  $I$ . For all  $i \in I$ , let  $J_i$  be also a directed set and consider a net  $(x_j^i)_{j \in J_i}$  that  $\mathcal{C}$ -converges to  $x^i \in X$ . Assume also that  $x^i \xrightarrow{i \in I}^{\mathcal{C}} x \in X$ . Denote  $F := I \times \prod_{i \in I} J_i$ . The set  $J$  is directed by the pointwise order inherited from  $I$  and  $J_i$ 's. We then require that the following convergence must hold:

$$x_{f(i)}^i \xrightarrow{(i, f) \in F}^{\mathcal{C}} x.$$

**Theorem 1.1.** (J. L. Kelley, see [1]): Let  $X$  be a set with a convergence class  $\mathcal{C}$ . For a set  $A \subset X$ , define  $\overline{A}$  to be the set of all  $x \in X$  such that there is a net  $(x_i)_{i \in I} \subset A$  that  $\mathcal{C}$ -converges to  $x$ . Then map  $A \mapsto \overline{A}$  is a closure operator that endows  $X$  with a topology in which a net  $(x_i)_{i \in I} \subset X$  converges to  $x \in X$  iff it  $\mathcal{C}$ -converges to  $x$ .

## 2. The topology of synchronous convergence

**Theorem 2.1.** Let  $X$  and  $Y$  be two convergence spaces, and let  $\mathcal{C} = \mathcal{C}(X, Y) \subset Y^X$  be the family of all continuous maps from  $X$  to  $Y$ . For a directed set  $I$ , a net  $(f_i)_{i \in I} \subset \mathcal{C}$ , and a point  $f \in \mathcal{C}$ , we say that  $f_i$  converges to  $f$  if and only if for every directed set  $J$  with a monotone final map  $\sigma : J \rightarrow I$ , and any net  $(x_j)_{j \in J} \subset X$ , we have

$$f_{\sigma(j)}(x_j) \xrightarrow{j \in J} f(x) \in Y.$$

We claim that the set  $\mathcal{S}$  consisting of pairs  $((f_i)_{i \in I}, f)$  such that  $f_i \xrightarrow{i \in I} f$  as defined above, is a convergence class on  $\mathcal{C}$ , and subsequently provides a topology on  $\mathcal{C}$ , called the topology of **semi-uniform convergence**.

*Proof:* We first need to verify four axioms that a convergence class must satisfy:

1. Let  $I$  be a directed set, and let  $f \in \mathcal{C}$ . Since  $f$  is continuous, for any directed set  $J$  and any net  $(x_j)_{j \in J} \subset X$  that converges to  $x \in X$ , we have  $f(x_j) \xrightarrow{j \in J} f(x) \in Y$ . Therefore the net  $(f)_{i \in I}$  converges to  $f$  in  $\mathcal{C}$ .
2. We need to show that any subnet of a convergent net converges to the same point. Let  $(f_i)_{i \in I} \subset \mathcal{C}$  be a net that converges to  $f \in \mathcal{C}$ , and let  $J$  be a directed set with  $\sigma : J \rightarrow I$ . To show that the subnet  $(f_{\sigma(j)})_{j \in J}$  converges to  $f$ , consider another directed set  $K$ , mapped to  $J$  via  $\tau : K \rightarrow J$ , and a net  $(x_k)_{k \in K} \subset X$  that converges to  $x \in X$ . Since the map  $\sigma \circ \tau : K \rightarrow I$  is monotone and final, by the definition of convergence we see that

$$f_{\sigma(\tau(k))}(x_k) \xrightarrow{k \in K} f(x),$$

which means that  $f_{\sigma(j)} \xrightarrow{j \in J} f$  in  $\mathcal{C}$ , as desired.

3. Let  $(f_i)_{i \in I} \subset \mathcal{C}$  be a net and assume that every subnet of  $(f_i)_{i \in I}$  has a subnet that converges to  $f \in \mathcal{C}$ . We need to show that the net  $(f_i)_{i \in I}$  itself converges to  $f$ . To that end, let  $J$  be a directed set with a map  $\sigma : J \rightarrow I$ , together with a net  $(x_j)_{j \in J} \subset X$  that converges to  $x \in X$ . Assume that the net  $(f_{\sigma(j)}(x_j))_{j \in J}$  does **not** converge to  $f(x)$  in  $Y$ . That means that there is a neighborhood  $U$  of  $f(x)$  and a subnet  $(f_{\sigma(\tau(k))}(x_{\tau(k)}))_{k \in K}$ , indexed by a directed set  $K$  with  $\tau : K \rightarrow J$ , such that  $f_{\sigma(\tau(k))}(x_{\tau(k)}) \notin U$  for all  $k \in K$ . At the same time, defining  $\alpha : K \rightarrow I$  by  $\alpha = \sigma \circ \tau$ , we see that  $(f_{\alpha(k)})_{k \in K}$  is a subnet of  $(f_i)_{i \in I}$ . Therefore, this subnet itself has a subnet that converges to  $f$ , i.e. there is a directed set  $L$  with a map  $\beta : L \rightarrow K$ , such that  $f_{\alpha(\beta(l))} \xrightarrow{l \in L} f \in \mathcal{C}$ . In particular, this implies that

$$f_{\sigma(\tau(\beta(l)))}(x_{\tau(\beta(l))}) \xrightarrow{l \in L} f(x).$$

But this is contradictory, since  $f_{\sigma(\tau(\beta(l)))}(x_{\tau(\beta(l))}) \notin U$  for all  $l \in L$ .

4. Finally, we need to show the diagonal property. Let  $(f_i)_{i \in I} \subset \mathcal{C}$  be a net converging to  $f \in \mathcal{C}$ . For every  $i \in I$ , consider also a directed set  $J_i$  and a net  $(f'_{(i,j)})_{j \in J_i} \subset \mathcal{C}$  that converges to  $f_i$ . Let

$$K := I \times \prod_{i \in I} J_i, \quad I \times J := \{(i, j) \mid i \in I, j \in J_i\},$$

with the preorder on  $K$  defined pointwise, and consider  $R : K \rightarrow I \times J$  defined by  $R(i, \alpha) = (i, \alpha(i))$ . We then need to show that the net

$$(f'_{R(i, \alpha)})_{(i, \alpha) \in K} \subset \mathcal{C}$$

converges to  $f$ . To that end, consider a directed set  $L$  with  $\tau : L \rightarrow K$  and a net  $(x_l)_{l \in L} \subset X$  converging to  $x \in X$ . We need to show the convergence

$$f'_{R(\tau(l))}(x_l) \xrightarrow[l \in L]{} f(x) \in Y.$$

Since the above convergence is topological, we can show it in the usual way. Let  $U$  be a neighborhood of  $f(x)$  in  $Y$ . We seek  $l_0 \in L$  such that

$$\forall l \geq l_0, f'_{R(\tau(l))}(x_l) \in U.$$

Consider the map  $\eta : L \rightarrow I$  defined by  $\eta(l) = i$ , where  $(i, \alpha) = \tau(l)$ . Clearly,  $\eta$  is a monotone final map, so by the definition of convergence,  $f_i \xrightarrow[i \in I]{} f$  implies that

$$f_{\eta(l)}(x_l) \xrightarrow[l \in L]{} f(x) \in Y.$$

Hence, there is  $l_0 \in L$  such that for all  $l \geq l_0$ , we have  $f_{\eta(l)}(x_l) \in U$ . Now, for any  $l$ , we have the convergence

$$f'_{(\eta(l), j)} \xrightarrow[j \in J_{\eta(l)}]{} f_{\eta(l)} \in \mathcal{C}.$$

Applying the definition of convergence for the constant net  $(x_l)_{j \in J_{\eta(l)}} \subset X$ , we see that

$$f'_{(\eta(l), j)}(x_l) \xrightarrow[j \in J_{\eta(l)}]{} f_{\eta(l)}(x_l) \in Y.$$

Therefore, for all  $l \geq l_0$ , there is  $j_0^l \in J_{\eta(l)}$  such that for all  $j \geq j_0^l$ , we have  $f'_{(\eta(l), j)}(x_l) \in U$ . Next, we define  $\alpha_0 \in \prod_{i \in I} J_i$  as follows:

$$\alpha_0(i) := \begin{cases} j_0^l, & \text{if } i = \eta(l) \text{ with } l \geq l_0, \\ \text{any } j \in J_i, & \text{otherwise.} \end{cases}$$

We then have  $(\eta(l_0), \alpha_0) \in K$ , and since  $\tau : L \rightarrow K$  is final, there is  $l_1 \in L$  such that  $l_1 \geq l_0$  and  $\tau(l_1) \geq (\eta(l_0), \alpha_0)$ .

Now, consider any  $l \geq l_1$ , with  $\tau(l) = (i, \alpha)$ . Since  $\tau(l) \geq \tau(l_1) \geq (\eta(l_0), \alpha_0)$ , we see that  $\alpha \geq \alpha_0$ , and so  $\alpha(i) \geq \alpha_0(i)$  for all  $i \in I$ . Since  $l \geq l_0$  and  $\alpha(\eta(l)) \geq \alpha_0(\eta(l)) = j_0^l$ , we have

$$f'_{R(\tau(l))}(x_l) = f'_{(\eta(l), \alpha(\eta(l)))}(x_l) \in U.$$

Since the neighbourhood  $U$  was arbitrary, this implies the convergence

$$f'_{R(\tau(l))}(x_l) \xrightarrow[l \in L]{} f(x) \in Y,$$

as desired.

This shows that our notion of convergence on  $\mathcal{C}$  is a convergence class, and hence the closure operator defined by

$$\overline{E} := \left\{ f \in \mathcal{C} \mid \exists (f_i)_{i \in I} \subset E \text{ st } f_i \xrightarrow[i \in I]{} f \right\}$$

endows  $\mathcal{C}$  with a topology in which convergence coincides with the one defined by  $\mathcal{S}$ . ■

**Proposition 2.1.** *Let  $X$  and  $Y$  be topological spaces and assume that  $X$  is first countable. Construct the topology on  $\mathcal{C} = \mathcal{C}(X, Y)$  as above. Then a sequence  $\{f_k\}_{k=1}^\infty \subset \mathcal{C}$  will converge to  $f \in \mathcal{C}$  if and only if  $f_k(x_k) \rightarrow f(x)$  in  $Y$  whenever  $x_k \rightarrow x$  in  $X$ .*

*Proof:* Consider a sequence  $\{f_k\}_{k=1}^\infty \subset \mathcal{C}$ . Suppose that it converges to a point  $f \in \mathcal{C}$ . Note that this sequence is a net with  $I = \mathbb{N}$ . Taking  $J = I = \mathbb{N}$  and  $\sigma = \text{id}$ , we see that for any sequence  $\{x_k\}_{k=1}^\infty \subset X$  converging to  $x \in X$ , we have

$$f_k(x_k) \xrightarrow[k \rightarrow \infty]{} f(x). \tag{1}$$

Conversely, suppose that  $f_k(x_k) \rightarrow f(x)$  whenever  $x_k \rightarrow x \in X$ . Let us first see that for any subsequence  $\{f_{k_l}\}_{l=1}^{\infty}$  and any sequence  $x_l \rightarrow x \in X$ , we have  $f_{k_l}(x_l) \rightarrow f(x) \in Y$ . Indeed, given such sequences, for any  $k \in \mathbb{N}$  we define  $\alpha(k)$  to be the minimum of all  $l \in \mathbb{N}$  such that  $k \leq k_l$ . Then the sequence  $\{x_{\alpha(k)}\}_{k=1}^{\infty}$  converges to  $x$ , because if  $x_l \in V \ni x$  for all  $l \geq l_0$ , then for all  $k \geq k_{l_0}$ , we have  $\alpha(k) \geq \alpha(k_{l_0}) = l_0$ , and so  $x_{\alpha(k)} \in V$ .

Now, since  $\alpha(k_l) = l$  for all  $l \in \mathbb{N}$ , the sequence  $\{f_{k_l}(x_l)\}_{l=1}^{\infty}$  is a subsequence of  $\{f_k(x_{\alpha(k)})\}_{k=1}^{\infty}$ . And since the latter converges to  $f(x)$  by (1), so does the former.

Finally, we show that  $f_k \xrightarrow[k \rightarrow \infty]{} f \in \mathcal{C}$ . Let  $J$  be a directed set with a monotone final map  $\sigma : J \rightarrow \mathbb{N}$ , and let  $(x_j)_{j \in J} \subset X$  be a net that converges to  $x \in X$ . We need to show that the net  $(f_{\sigma(j)}(x_j))_{j \in J}$  converges to  $f(x)$ . Assume the contrary. Then we have a neighborhood  $U$  of  $f(x)$  and a subnet  $(f_{\sigma(\tau(l))}(x_{\tau(l)}))$ , with  $\tau : L \rightarrow J$ , such that

$$\forall l \in L, f_{\sigma(\tau(l))}(x_{\tau(l)}) \notin U. \quad (2)$$

Since  $X$  is first countable, since the net  $(x_{\tau(l)})_{l \in L}$  converges to  $x$ , we can find a sequence  $\{l_i\}_{i=1}^{\infty}$  such that the sequence  $\{x_{\tau(l_i)}\}_{i=1}^{\infty}$  converges to  $x$ . Furthermore, by ensuring that  $\sigma(\tau(l_{i+1})) > \sigma(\tau(l_i))$ , we can define  $k_i := \sigma(\tau(l_i))$  and obtain a subsequence  $\{f_{k_i}\}_{i=1}^{\infty}$  of  $\{f_k\}_{k=1}^{\infty}$ . As shown before, we then have

$$f_{\sigma(\tau(l_i))}(x_{\tau(l_i)}) \xrightarrow{i \rightarrow \infty} f(x),$$

which contradicts (2). It then follows that  $f_k \xrightarrow[k \rightarrow \infty]{} f$ , as desired. ■

**Proposition 2.2.** *If  $X$  is a topological space and  $Y$  is a TVS, then the space  $\mathcal{C}(X, Y)$ , with the topology of semi-uniform convergence and pointwise linear structure, is also a TVS.*

| *Proof:* Follows easily from the definition of the topology on  $\mathcal{C}(X, Y)$  and the fact that  $Y$  is a TVS. ■

**Definition 2.1.** Let  $X$  be a first countable topological vector space, and let  $n \in \mathbb{N}_0$ .

The *curried function set*  $H_n(X)$  is defined as follows:

- If  $n = 0$ , we set  $H_0(X) := X$ ;
- If  $H_n(X)$  is defined, we define  $H_{n+1}(X)$  to be the set of all functions from  $X$  to  $H_n(X)$ .

The *curried continuous space*  $C_n(X)$  is a TVS, defined recursively:

- If  $n = 0$ , we set  $C_0(X) := X$ ;
- If  $C_n(X)$  is defined, we set  $C_{n+1}(X) := \mathcal{C}(X, C_n(X))$ , with pointwise linear structure and the topology of semi-uniform convergence.

**Remark 2.1.** Any function  $f \in C_n(X)$  naturally corresponds to a multivariable function  $\tilde{f} : X^n \rightarrow X$ :

$$\tilde{f}(x_1, x_2, \dots, x_n) = f(x_1)(x_2) \dots (x_n).$$

**Lemma 2.1.** *For all  $n \in \mathbb{N}$ , a function  $f \in H_n(X)$  lies in  $C_n(X)$  if and only if the corresponding function  $\tilde{f} : X^n \rightarrow X$  is continuous.*

*Proof:* First assume that  $f \in C_n(X)$ . To show that  $\tilde{f}$  is continuous, consider a sequence  $x_k^{(n)} \rightarrow x_0^{(n)} = (x_0^1, x_0^2, \dots, x_0^n) \in X^n$ . Since  $f$  is continuous, we see that  $f(x_k^1) \rightarrow f(x_0^1)$ . If  $n = 1$ , we are done. Otherwise, the convergence  $f(x_k^1) \rightarrow f(x_0^1)$  occurs in the topology of semi-uniform convergence, and so by [Proposition 2.1](#) we have

$$f(x_k^1)(x_k^2) \rightarrow f(x_0^1)(x_0^2).$$

Repeating this argument as needed, we obtain

$$f(x_k^1)(x_k^2) \dots (x_k^n) \rightarrow f(x_0^1)(x_0^2) \dots (x_0^n),$$

which is to say that  $\tilde{f}(x_k^{(n)}) \rightarrow \tilde{f}(x_0^{(n)})$ , as desired.

We will prove the converse by induction.

- If  $n = 1$ , then the statement is trivial.
- Suppose the statement is proven for  $C_n(X)$ . Consider  $f \in H_{n+1}(X)$  such that  $\tilde{f}: X^{n+1} \rightarrow X$  is continuous. By the induction hypothesis, we know that  $f(x) \in C_n(X)$  for all  $x \in X$ , so  $f$  is a function from  $X$  to  $C_n(X)$ . It remains to show that  $f$  is continuous. This is done in the same way as the first part of our proof.

■

## Bibliography

[1] J. L. Kelley, *General Topology*, vol. 27. Graduate Texts in Mathematics, 1975.